

ConstructHub

Software Requirements Specification

Construction Management System ERP

Team: 25_ConstructHub

Domain: Construction Management System ERP

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1. Functional Requirements

Functional requirements describe what the system is expected to do, including the services it provides and the actions it performs in response to user interactions.

A. User & Role Management

- Users can access the system based on assigned roles
- Different roles include:
 - Project Manager
 - Site Engineer
 - Finance Manager
 - Client
- Each role has restricted access based on responsibilities

B. Task Management

- Project Manager can create construction tasks
- Define task priority, start date, and deadline
- Assign tasks to Site Engineers
- Modify or reassign tasks if required
- Mark tasks as completed after execution

C. Site Reporting

- Site Engineers can submit periodic site reports
- Reports include progress details, remarks, and issues
- System validates and stores submitted reports
- Project Manager is notified when reports are submitted

D. Task Status Tracking

- Site Engineers can update task status
- Record task completion percentage
- Add status remarks
- System updates task history and dashboard

E. Project Schedule Management

- Project Manager can create project schedules
- Define milestones and dependencies
- Monitor schedule progress
- Revise schedules in case of delays

F. Budget and Billing Management

- Finance Manager can configure approved project budgets
- Record and track project expenses
- Monitor remaining budget
- Generate bills for completed work

G. Payment Approval

- Client can view pending bills
- Client can approve or reject bills
- System records approval decisions
- Finance Manager is notified of approval status
- System does not process payments directly

H. Reporting & Monitoring

- View dashboard showing:
 - Task progress
 - Schedule status
 - Budget utilization
 - Billing status
- Dashboard is read-only for Clients
- Feedback can be submitted through the dashboard

2. Non-Functional Requirements

Non-functional requirements specify quality attributes and constraints of the system such as performance, security, and reliability.

A. Product Requirements

1. Efficiency Requirements

- System should handle multiple users concurrently
- Task and report updates should be processed efficiently

2. Performance Requirements

- Common user actions should respond within acceptable time
- Dashboard data should load without noticeable delay

3. Space Requirements

- Efficient database storage for project, task, and report data
- Optimize storage for documents and images

4. Usability Requirements

- Simple and intuitive interface for non-technical users
- Minimal training required
- Consistent navigation across roles

5. Dependability Requirements

- System should be reliable during working hours
- Prevent loss of project and financial data
- Support data backup and recovery

6. Security Requirements

- Secure login and authentication mechanisms
- Role-based access control
- Protect sensitive project and financial information

B. Organizational Requirements

1. Operational Requirements

- System should operate on web platforms
- Accessible using standard web browsers

2. Environmental Requirements

- Should work under moderate network conditions

- Handle temporary connectivity issues gracefully

3. Development Requirements

- Modular design for easy maintenance
- Support future enhancements without major redesign

C. External Requirements

1. Regulatory Requirements

- Follow standard construction project management practices
- Maintain audit logs for critical actions

2. Ethical Requirements

- Ensure transparency in task assignment and billing
- Prevent unauthorized data manipulation

3. Legislative Requirements

- Comply with applicable IT and data protection laws

4. Accounting Requirements

- Track expenses and billing records accurately
- Maintain financial reports for auditing

5. Safety / Security Requirements

- Protect system from unauthorized access
- Ensure integrity of project and financial data

3. UML Diagrams

The following UML diagrams illustrate the system's use cases, activity flows, and sequence interactions across all user roles within the ConstructHub platform.

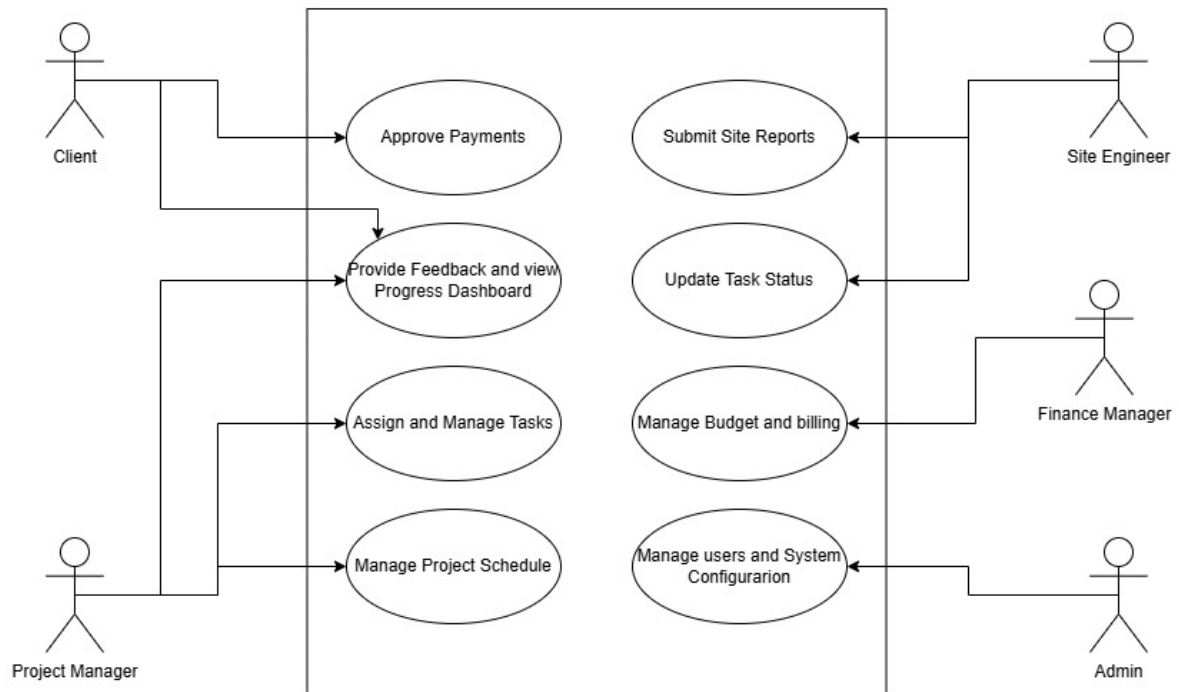


Figure 1: Use Case Diagram

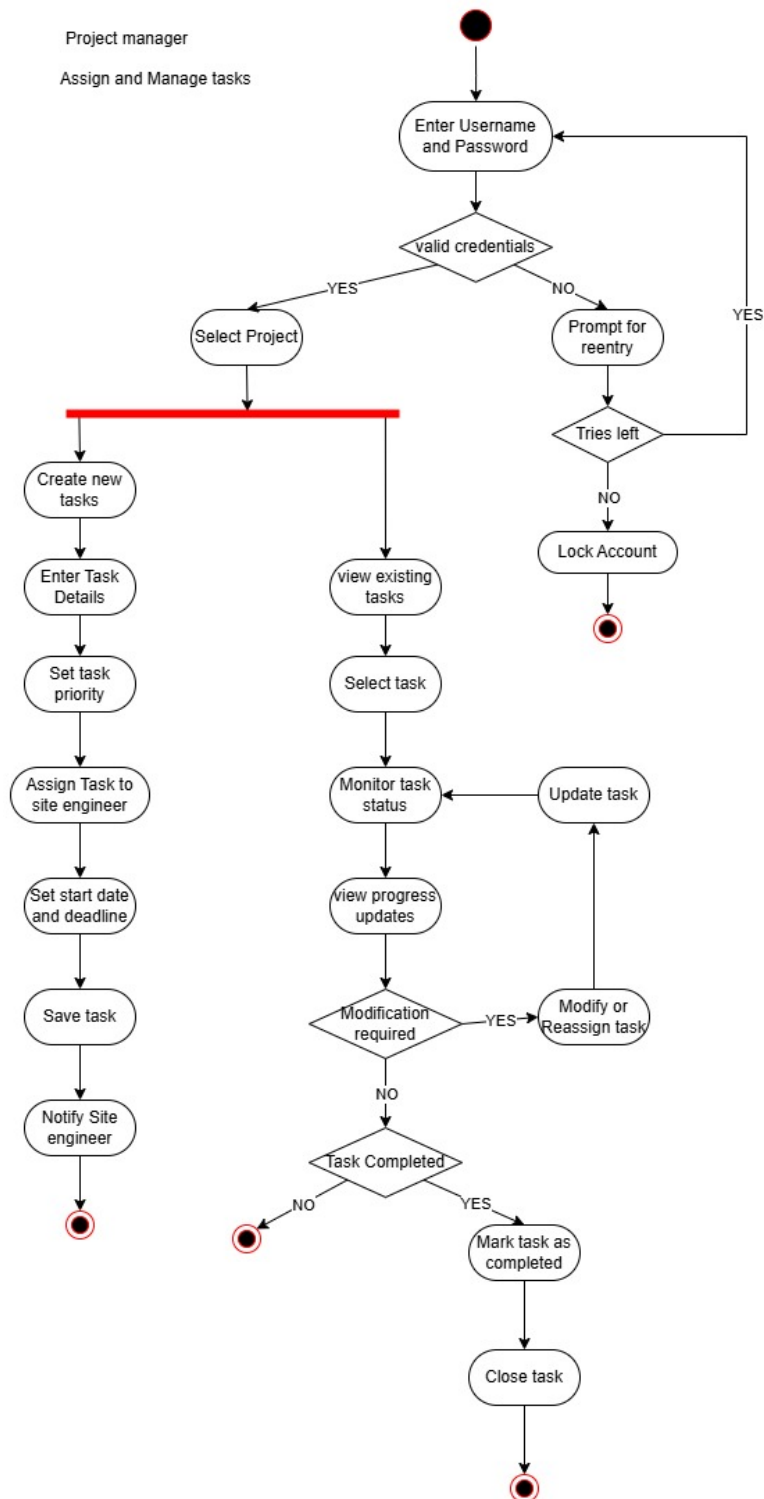


Figure 2: Activity Diagram – Assign and Manage Tasks (Project Manager)



Figure 3: Activity Diagram – Update Task Status (Site Engineer)

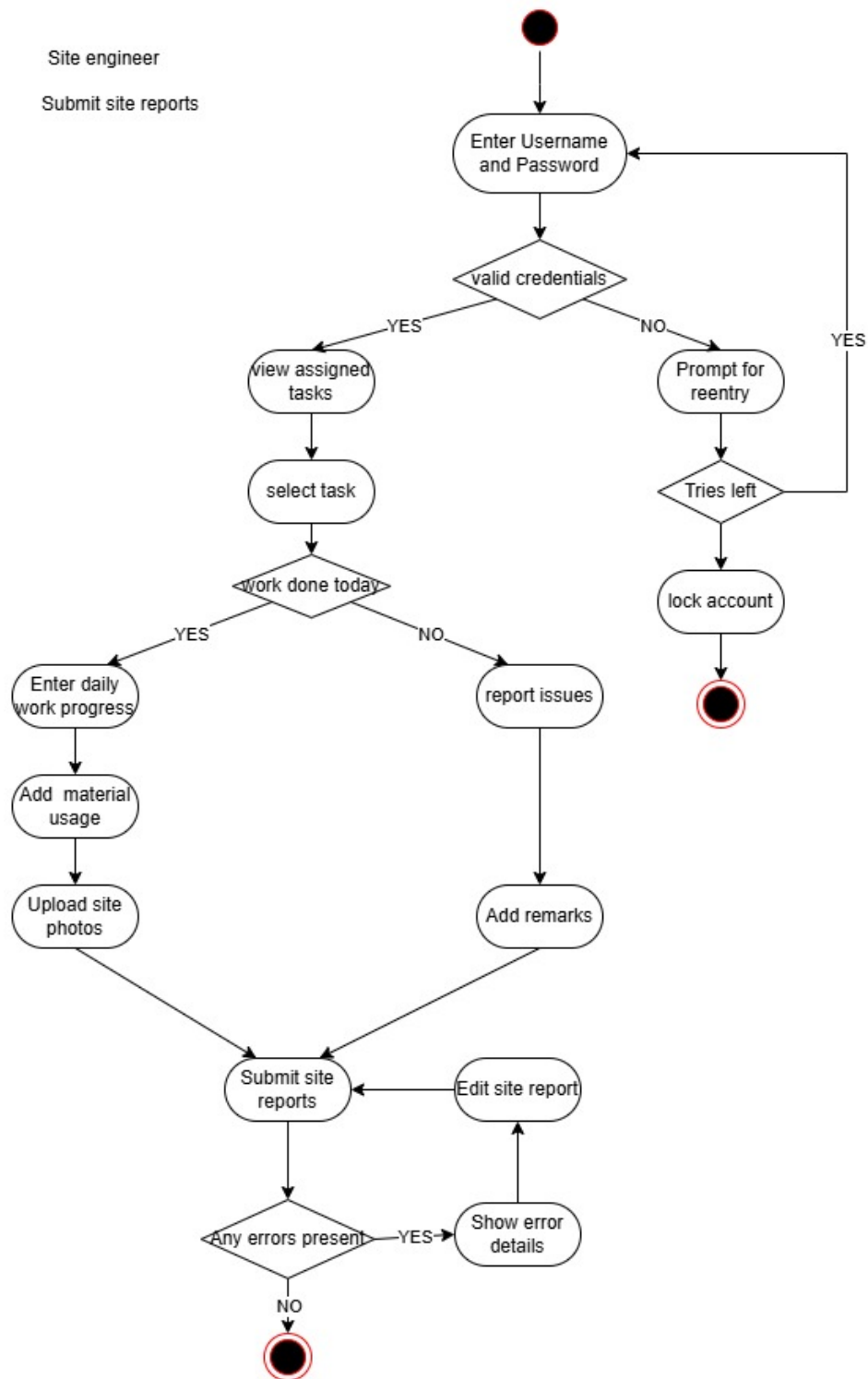


Figure 4: Activity Diagram – Submit Site Reports (Site Engineer)

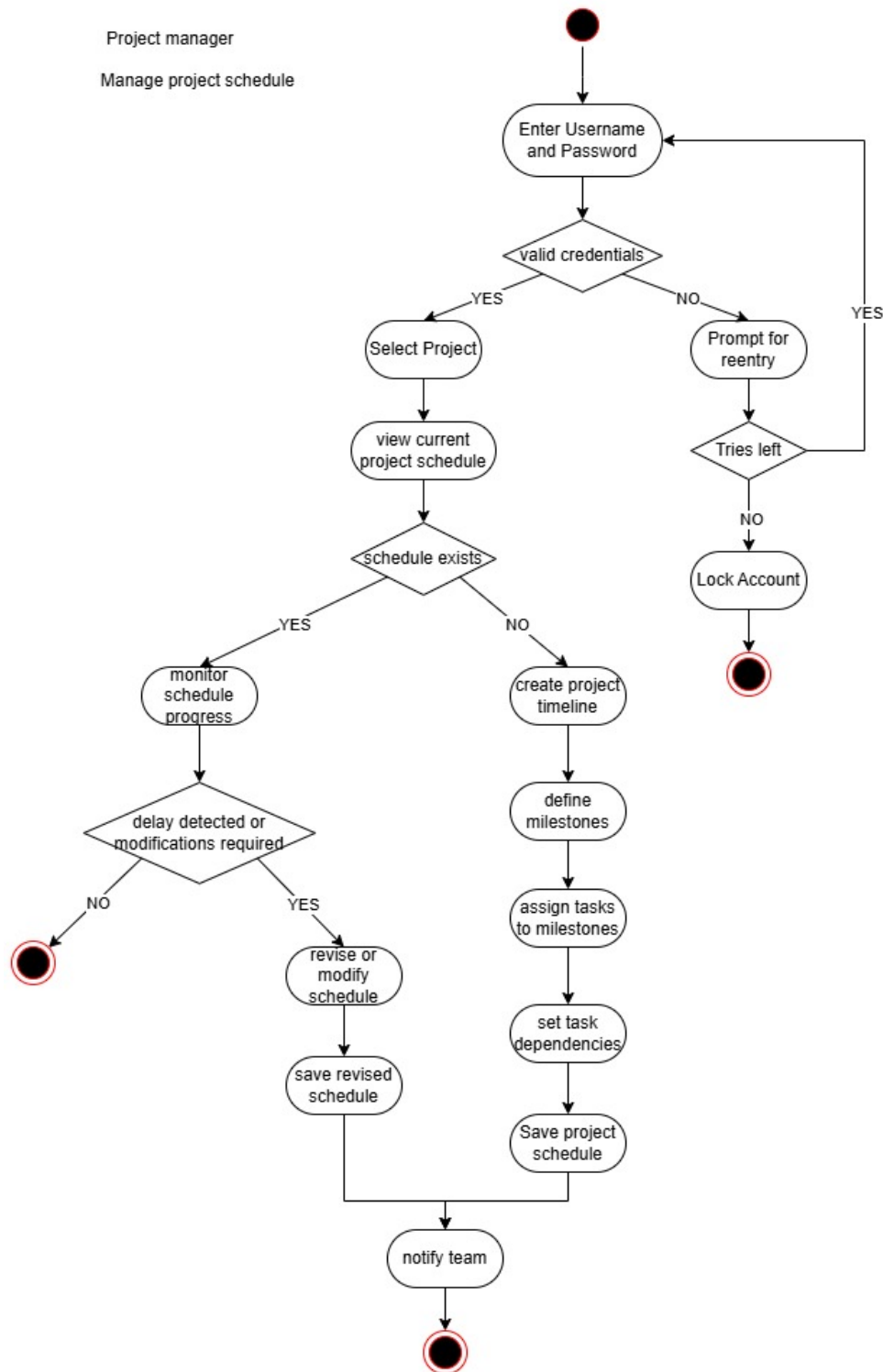


Figure 5: Activity Diagram – Manage Project Schedule (Project Manager)

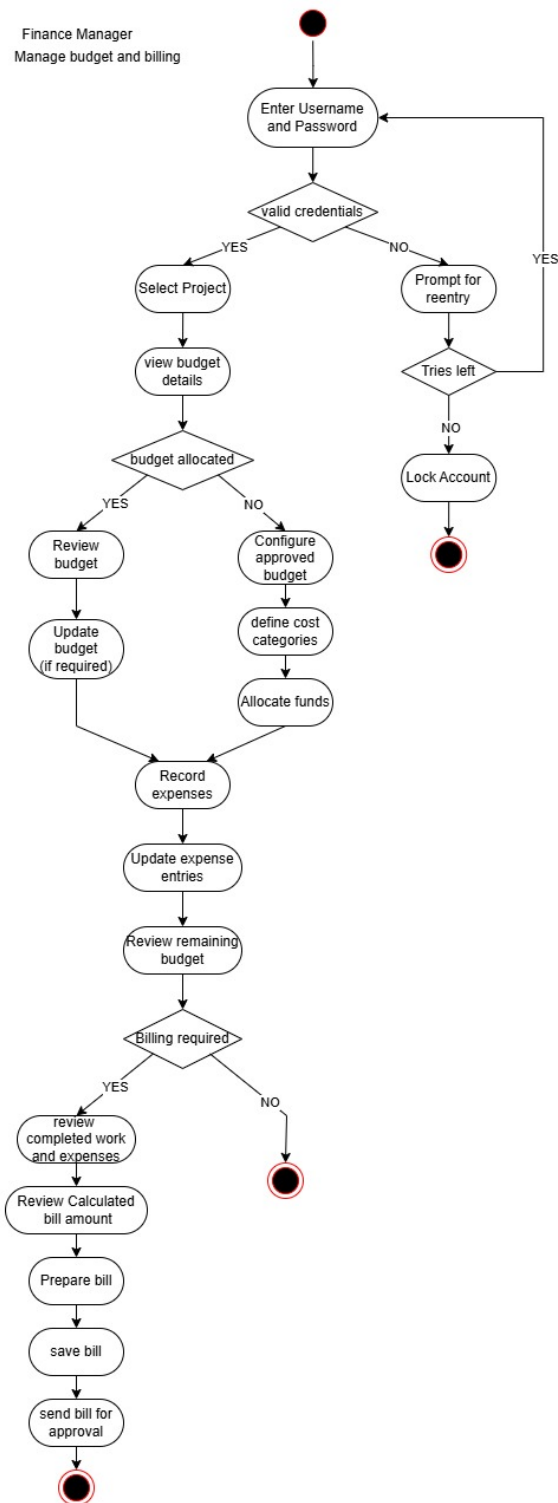


Figure 6: Activity Diagram – Manage Budget and Billing (Finance Manager)

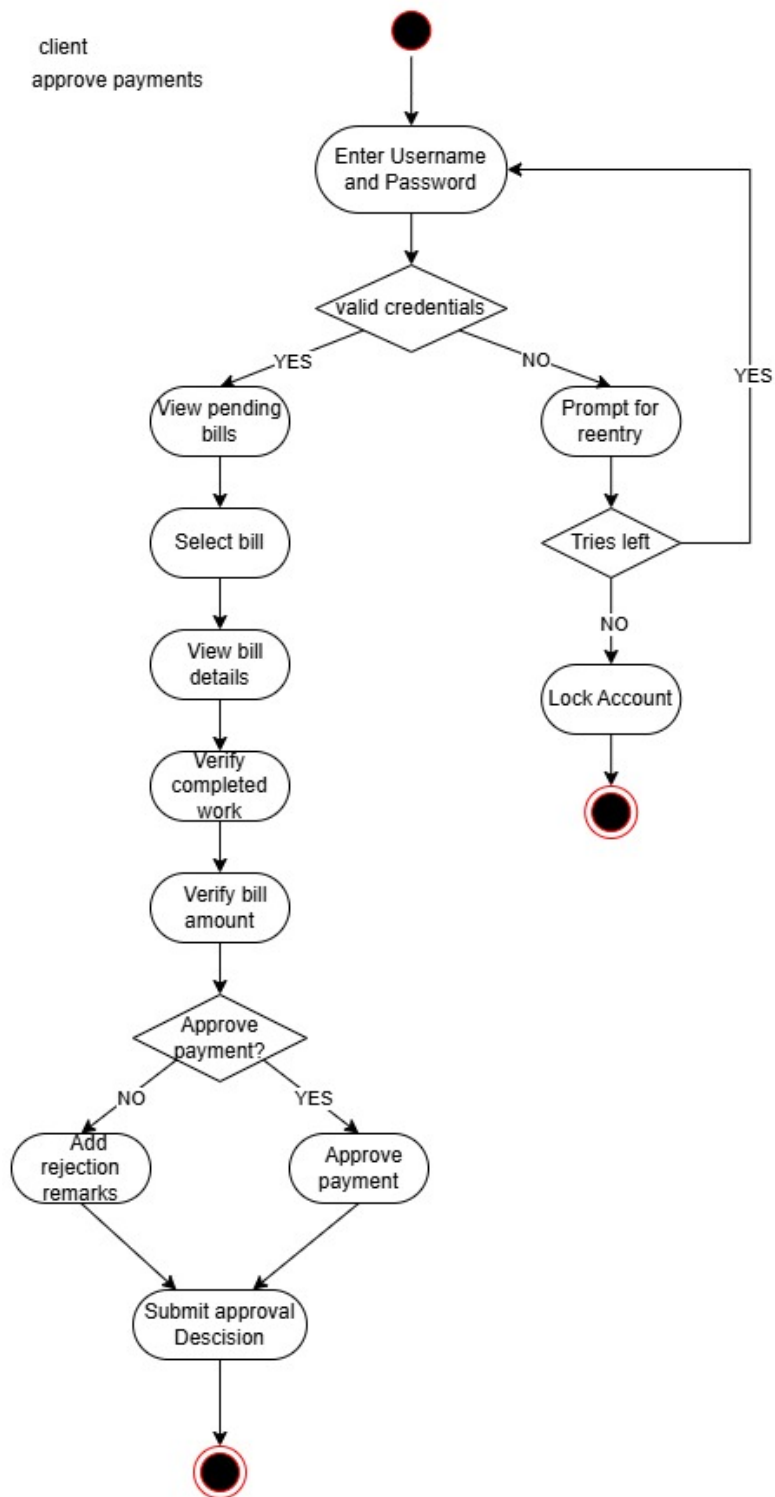


Figure 7: Activity Diagram – Approve Payments (Client)

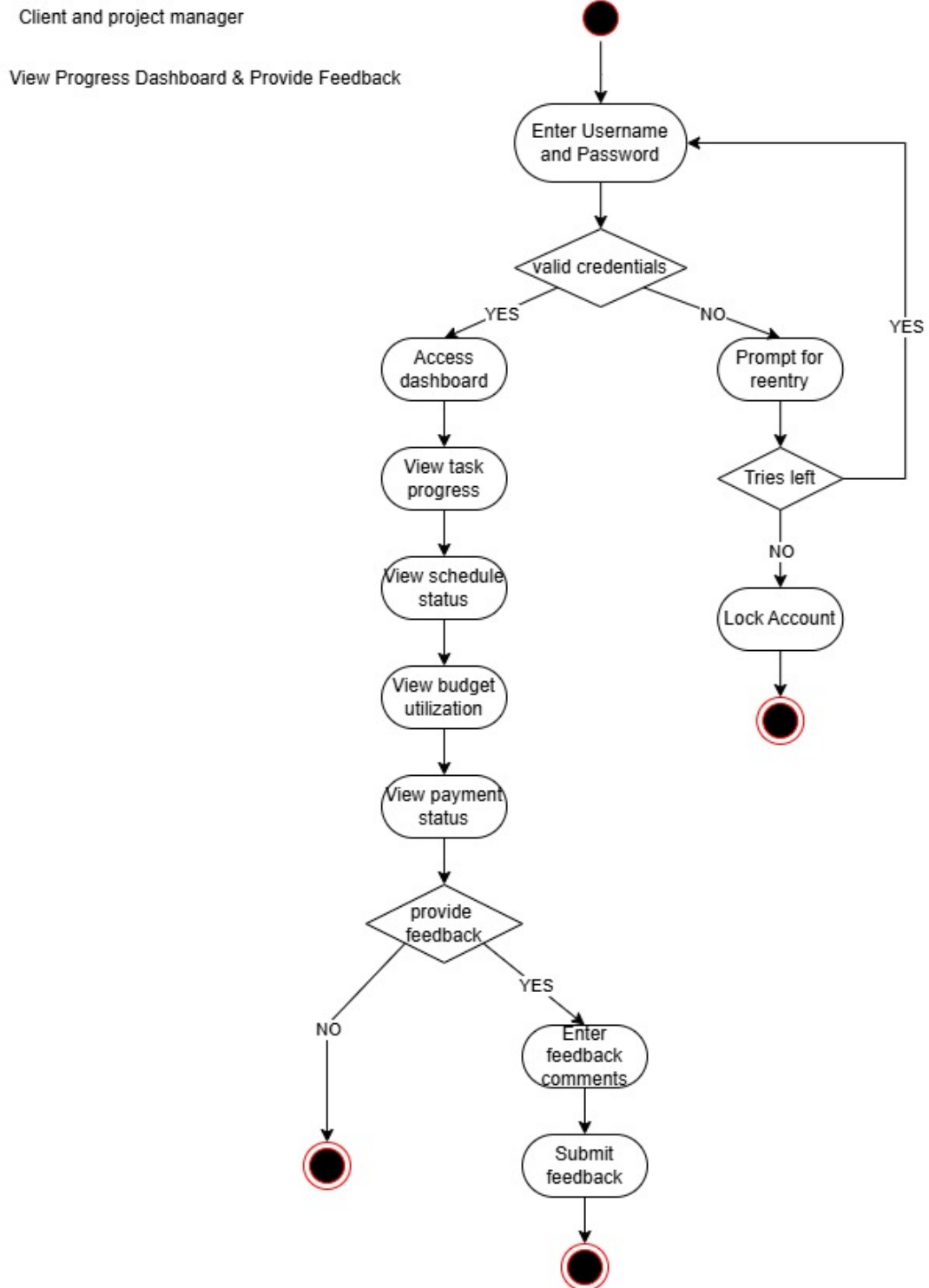


Figure 8: Activity Diagram – View Progress Dashboard & Provide Feedback

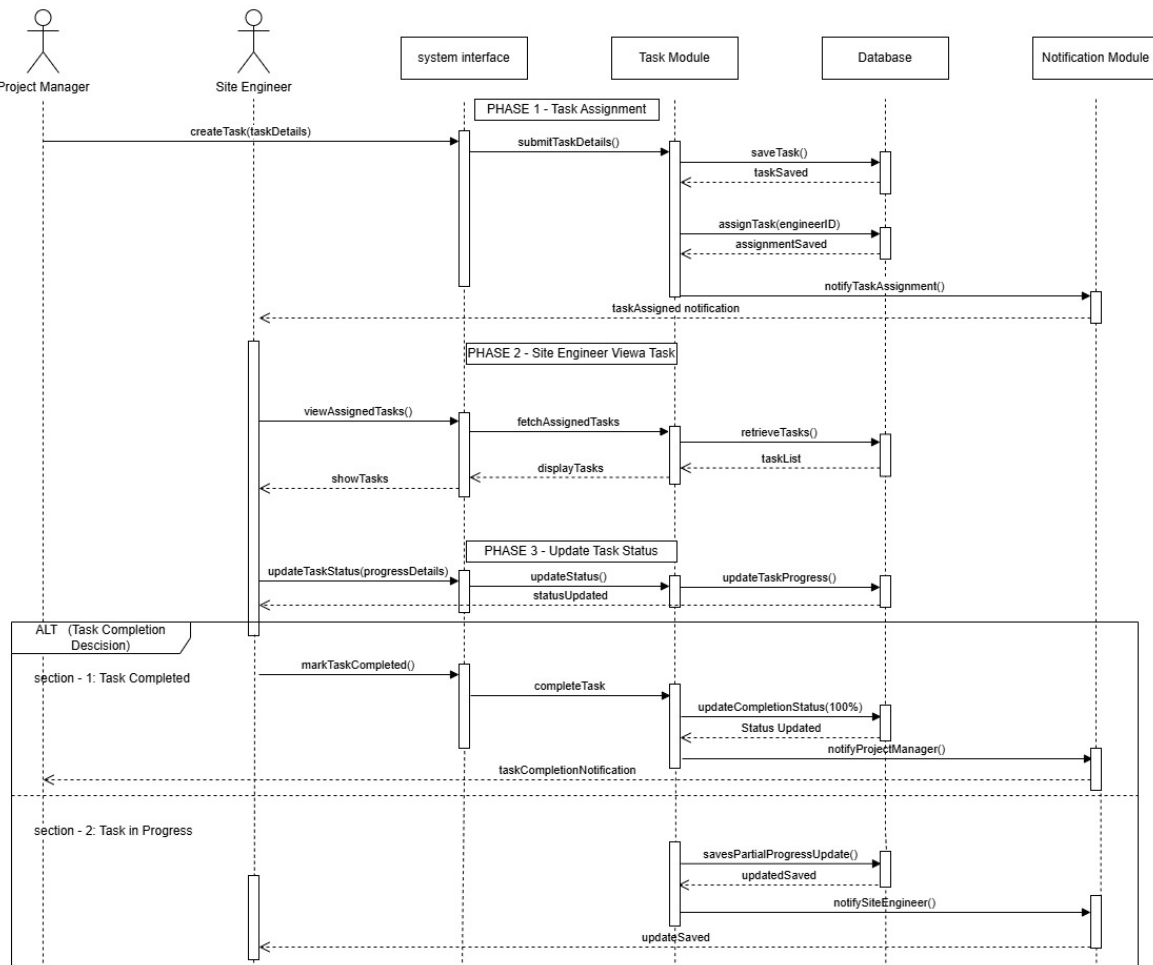


Figure 9: Sequence Diagram – Manage and Assign Tasks

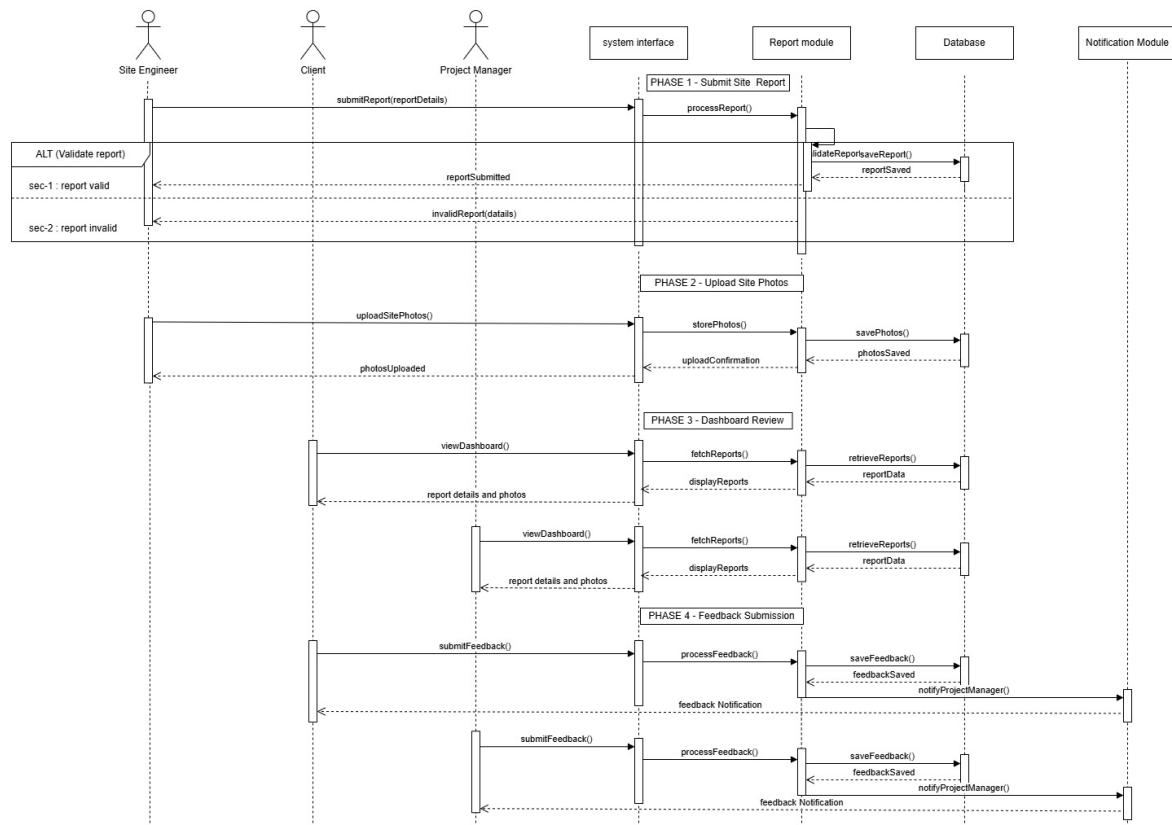


Figure 10: Sequence Diagram – Site Report Submission and Feedback

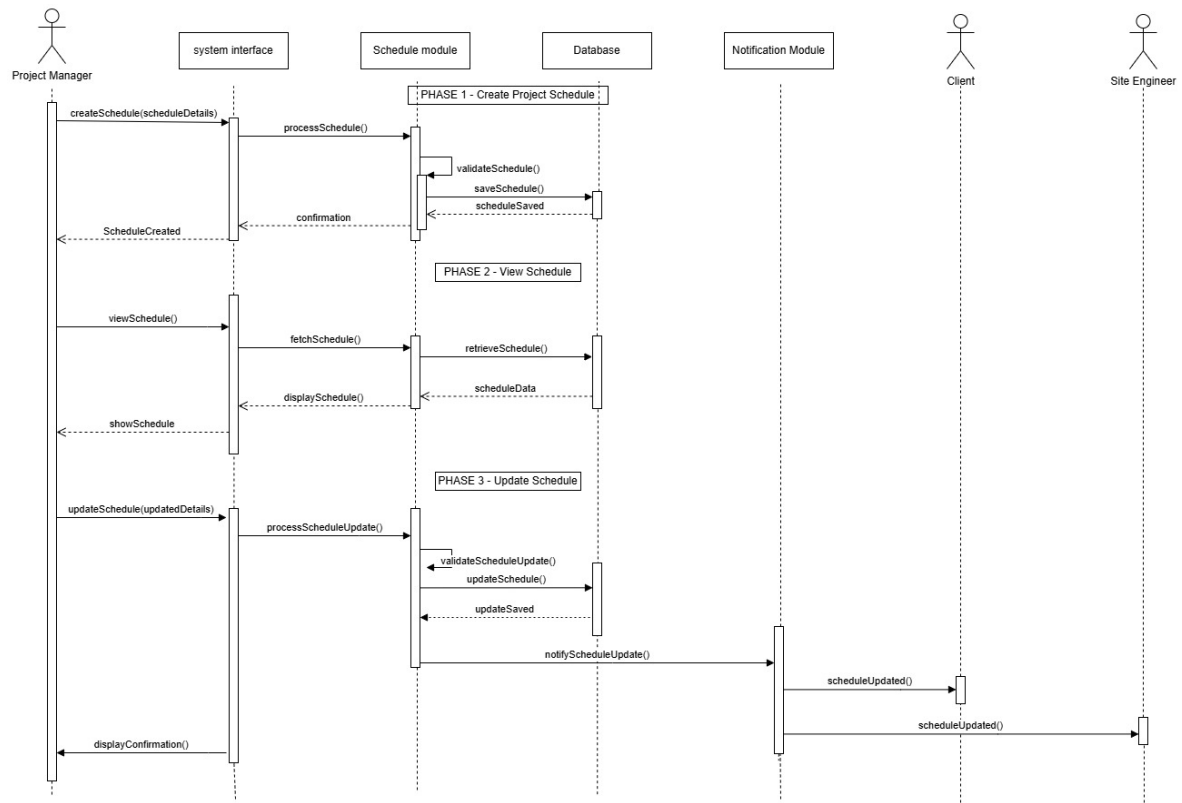


Figure 11: Sequence Diagram – Project Schedule Management

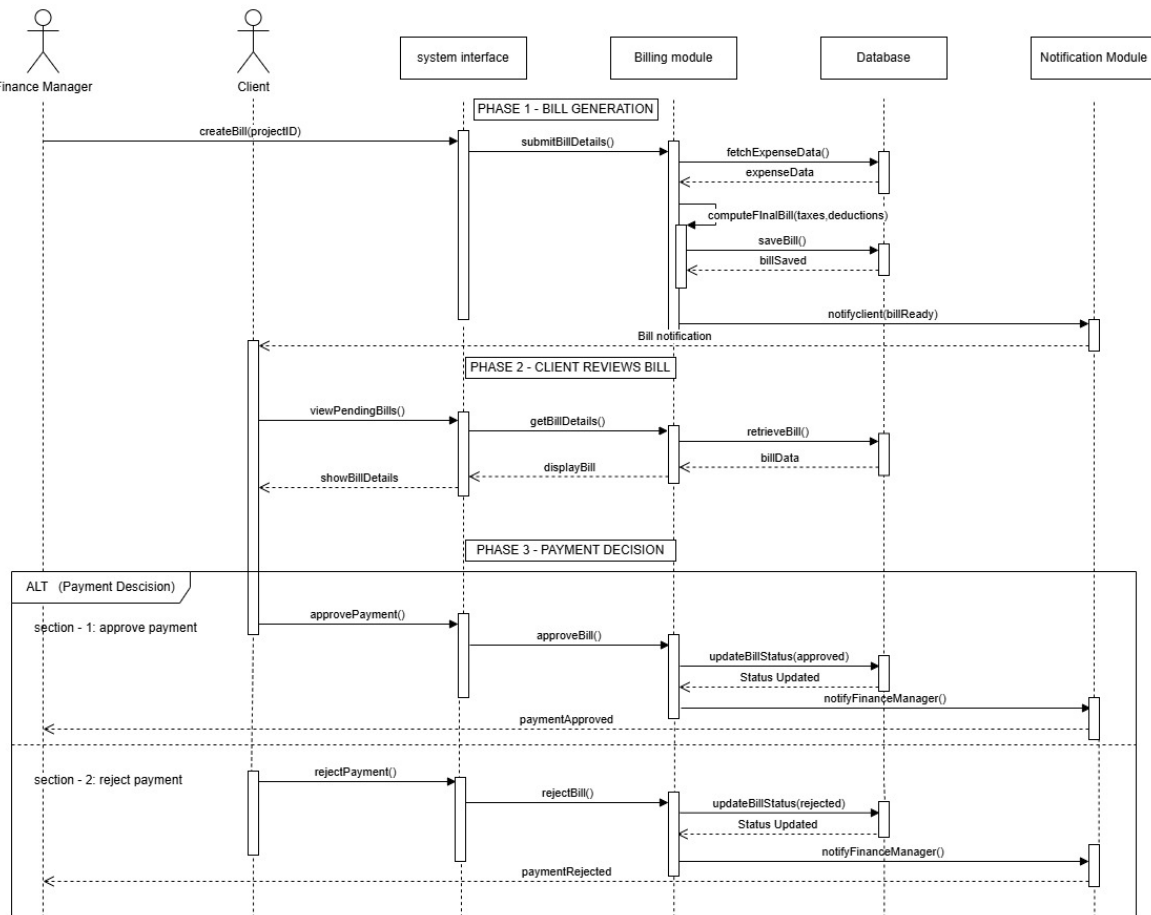


Figure 12: Sequence Diagram – Billing and Payment Approval